



XLR-KGDD: leveraging LLM and RAG for knowledge graph-based explainable disease diagnosis using multimodal clinical information

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Abstract

Large Language Models (LLMs) are making a big impact in Artificial Intelligence due to their ability to perform tasks as humans. However, using LLMs in many domain-specific tasks is a relatively unexplored area, specifically in disease diagnosis. This is due to challenges such as multiple modalities in patient clinical information and LLM's high memory and computational power requirements. This study proposes a framework, XLR-KGDD, that overcomes these challenges and performs an LLM-based disease diagnosis. Additionally, XLR-KGDD generates explanations to establish the trust of clinicians and support the diagnosis. These explanations are precise and unambiguous as they adhere to the standard medical guidelines and are presented in natural language. The proposed framework maps multimodal patient clinical information to a patient Knowledge Graph (KG) using the N2K mapper and CheXzero. Prompt Engineering is then applied to create an LLM-compatible input prompt from the patient KG. The framework employs a Parameter Efficient Fine-tuning technique to fine-tune LLM efficiently by optimizing numerical computations and memory requirements. The framework uses Retrieval Augmented Generation to provide standard medical guidelines as context to the LLM, addressing the issue of hallucinations in LLMs and generating coherent explanations. A system based on the XLR-KGDD framework was developed and tested on the multimodal MIMIC-Eye dataset. The LLaMA-3 LLM shows an AUC value of 0.88 and 0.91 in ROC and PR curves, respectively, in diagnosing patients with CHF disease. Furthermore, the system-generated explanations support the diagnosis with evidence from medical guidelines.

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1 Introduction

Large Language Models (LLMs) have been a trending topic since the emergence of OpenAI's Chat Generative Pre-trained Transformer (GPT) platform [1]. LLMs have been trained on large datasets that enable them to interact with users in natural language and provide responses based on their queries. Following the introduction of ChatGPT by OpenAI, other big companies such as Google and Meta have also launched their LLMs. Google launched its PaLM [2] LLM in 2023, while Meta launched its LLM series, LLaMA. Largely, LLMs seem to work well on general Question-Answering tasks [3, 4], especially in the medicinal domain, there are a number of Clinical ChatBots [5–7]. However, their application in other downstream tasks, such as disease diagnosis, is an open research area that is relatively untouched.

Some existing works focus on the application of LLMs in downstream tasks [8–10]. However, these works primarily focus on textual data inputs, such as sentiment classification [11] and document classification [8]. In the real world, the data is present in varied formats. For instance, in an Electronic Health Record (EHR), symptoms are stored as text and radiology images in a dicom format. Clinical decisions, such as disease diagnosis, are made after analyzing these multi-modular patient attributes [12]. This work explores the potential of LLMs in disease diagnosis with multimodal data.

The paper proposes a framework, XLR-KGDD that takes multimodal patient data as input to diagnose disease with explanations using Knowledge Graph (KG), LLM, and Retrieval Augmented Generation (RAG). KG is a knowledge representation technique that portrays complex relations of real-world domains in a machine and human-understandable format. The framework leverages KG to represent clinical multimodal data in triples which are further used to create the input prompt using prompt engineering. The LLM is fine-tuned using the created prompts for disease diagnosis. Also, to ensure the adherence of diagnosis to the standard medical guidelines and enhance the trust of medical professionals, explanations are generated for the given diagnosis by referring to medical guidelines.

However, the following challenges are faced in this study which are addressed in the proposed framework. The LLMs face three major challenges: a lack of openly accessible models, hardware requirements, and hallucinations. Though the initial version of ChatGPT was openly available, the advanced versions of OpenAI's LLMs have been commercialized for monetary benefits. Hence, it is not easily accessible for research purposes. Along similar lines, PaLM and PaLM2 are two LLMs developed by Google that have not been released openly. The unavailability of open-source LLMs presents a challenge for LLM researchers in realizing the full potential of LLMs.

The size of LLMs is a double-edged sword for researchers as it presents both a benefit and a problem. LLMs are trained on voluminous data and contain billions of parameters. As a result, they perform tasks efficiently and with high accuracy. However, loading a billion-parameter model on a non-commercial machine is challenging as it needs a large memory for loading and processing numerical computations. Additionally, in a general query response generation, the LLMs suffer from the problem of hallucinations [13]. It may either generate irrelevant inferences to the user's query or an invalid response in case of insufficient training

corpus. This paper addresses these challenges in a downstream task of clinical decision, particularly disease diagnosis.

The contributions of the paper are as follows.

1. This paper proposes a novel disease diagnosis framework using Knowledge Graph (KG), LLM, and Retrieval Augmented Generation (RAG). It generates disease diagnosis supported with human interpretable explanations based on multimodal clinical data of EHR.
2. This paper applies prompt engineering to generate input prompts (disease diagnosis prompt and explanation prompt) for the LLMs from semantically mapped patient KG.
3. The challenges concerning high memory and computational requirements are addressed in the paper by using a Parameter Efficient Fine-Tuning (PEFT) technique called QLoRA. It fine-tunes different LLMs (LLaMA-3, LLaMA-2, and Mistral) to diagnose diseases and generate explanations.
4. The framework uses the RAG technique to generate diagnosis explanations adhering to standard medical guidelines. RAG leverages external knowledge sources (medical guidelines) to provide a context to LLM and generates relevant inferences. It helps to overcome the problem of hallucination.

The organization of the paper is as follows. Section 2 discusses the literature on LLMs and their applications in downstream tasks, especially in medicinal domains. Section 3 presents the proposed framework and a detailed explanation of its components. An experimental study conducted on the MIMIC-Eye dataset is described in Sect. 4. Section 5 discusses the results. Section 6 concludes the paper.

2 Literature survey

This section discusses various LLMs and their variants that are available in the literature. It also presents the applications of these LLMs in various downstream tasks, especially in the medicinal domain.

Large Language Models (LLMs): The path of LLMs was paved with the evolution of OpenAI's GPT [14] in 2018. However, it largely gained public attention with ChatGPT-3 [1], which takes input in natural language and generates output in the same pattern. ChatGPT-3 was trained on a large and varied dataset that generates relevant responses on general topics. Its successor, ChatGPT-4 [15], introduced in 2023, was much larger than its predecessor. There exist other LLMs besides GPTs which are discussed below.

The authors in [16] have developed an intelligent communication system by leveraging the large textual dataset to interpret and generate human-relevant responses. They evaluated the performance of various LLMs by scaling the parameters from 10 Million (M) to 280 Billion (B). The model with 280 B parameters, named Gopher, gave the best performance. Following Gopher and GPT-3, Chinchilla [17] was another early LLM. The authors in [17] claimed a prominent statement about the relationship between the model size and training data. They concluded that if the size of a model increases twofold, then the required data for optimal model training also doubles. Following this principle, the Chinchilla model with 70 B parameters was trained with 1.4 trillion tokens. The Chinchilla model was trained with four times more tokens than the Gopher Model (with 280 B). It showed better performance than Gopher and GPT-3 with 175 B. Another LLM model, Pathways Language Model (PaLM), was introduced in the paper [2]. It is a 540 B parameter model trained using a Tensor Processing Unit (TPU)-based training approach called Pathways. This approach can efficiently train Deep Learning (DL) Models on multiple TPU Pods. The authors in [2] also concluded that

scaling LLMs produces better results in language understanding and generation tasks. In 2023, another series of LLMs was introduced by Meta, known as LLaMA [18], which is trained on publicly available datasets. These models are trained on millions of tokens, and the parameters of this series range from 7 to 65 B. The comparison in [18] illustrates that LLaMA with 13 B performs better than GPT-3 with 175 B and gives comparable results with Chinchilla with 70 B and PaLM with 540 B. Meta has openly released the pre-trained weights of the LLaMA series, which is accessible for further research. Other variants of LLaMA, such as LLaMA-2 and LLaMA-3, have been recently published. LLaMA-2 is an improved version of LLaMA with three updates. The LLaMA-2 has a 40% larger training corpus and increased length context length by twice compared to LLaMA. Also, the attention mechanism has been changed to grouped-query attention. LLaMA-3 has been trained on 15 trillion tokens and achieved better performance on a range of tasks. It makes the LLaMA series one of the best LLMs available in the market. Baize [19] is another LLM that generates responses based on the input dialogue of a user. The authors in [19] created a multi-turn chat dataset using ChatGPT to train the LLM. The base LLaMA model was fine-tuned using the Self-Distill technique with the customized dataset. The resultant model is a multi-turn chat LLM model.

LLMs in the medicinal domain: Generally, LLMs are used in textual-input-based applications such as Information Extraction (IE) [20], Question-Answering (QA) [21], Text Generation, and Natural Language Inference [22] in the medicinal domain. BioMedLM [4] is a small model compared to GPT-4 and Med-PaLM2 with 2.7 B parameters. It is an autoregressive model which is trained on PubMed articles and abstracts. BioMedLM generates multiple-choice QA results related to the biomedical domain. BioGPT [5] is a generative transformer model specifically trained on biomedical literature. It has been evaluated on six biomedical natural language tasks. The best performance of BioGPT is shown on PubMedQA tasks with 78.2% accuracy. The paper [23] introduced MedAlpaca, an open collection of datasets to train LLMs. A model based on LLaMA and trained on the MedAlpaca dataset has also been proposed in [23]. ChatDoctor [6] is another specialized medicinal language model for clinical advice. It uses a dataset with conversations between various patients and doctors from online consultation platforms containing around 1 lakh dialogues. The base LLaMA model was fine-tuned with the dialogue dataset. ChiMed-GPT [7] is a specialized LLM for Chinese medicine. It has a context length of 4096 tokens. The training process progressed in three parts: pre-training, followed by Supervised Fine-Tuning (SFT), and Reinforcement Learning with Human Feedback (RLHF). The ChiMedD-GPT was evaluated on IE, QA, and dialogue generation tasks.

Disease Diagnosis using LLMs: There are numerous disease diagnosis studies [24–26] using Machine Learning (ML) and DL algorithms. A recent survey [27] states that transformers perform better in various domains, including Natural Language Processing (NLP) and Computer Vision. Hence, a few LLM-based disease diagnosis studies from the literature are discussed here. Health-LLM [28] is a healthcare system that interacts with patients and uses LLM and ML to predict diseases. The system extracts symptoms and trains the LLM model to generate symptom descriptors. RAG in Health-LLM uses external medical resources to facilitate the generative process. Confidence scores are assigned to each symptom based on the similarity in medical documents using LLM and LLaMA-indexing. XGBoost is then used for disease classification. In [29], a Clinical Guidance Tree (CGT) was proposed that can be directly used by an LLM. It scraps flowcharts from medical literature in order to extract the clinical guidelines for diagnosis. These extracted flowcharts are processed further and converted to standard diagnostic decision trees. In [30], the identification and extraction of rare disease phenotypes have been explored. The phenotypic information of diseases is

generally present in an unstructured textual format. Due to this, identifying phenotypes is a major challenge with unannotated corpora. Hence, the authors have tried two techniques to analyze the phenotypes: fine-tuning with BioClinicalBERT and prompt engineering with ChatGPT. The use of ChatGPT achieved a higher F1-score of 0.778. In [31], the authors used various LLMs, including Gemini and GPT-3.5, in digital diagnostics based on patient symptoms. The Gemini model gave the best performance with 0.97 precision.

Explanations Generation: eXplainable AI (XAI) methods, such as Local Interpretable Model-Agnostic Explanations (LIME) and Shapley (SHAP) scores, are generally used to generate explanations [32]. XAI methods can be divided into two types: Model-Agnostic and Model-Specific. Model-Agnostic methods such as LIME, SHAP, and Partial Dependence Plot (PDP) are more popular than Model-Specific as they are not dependent on any ML model. However, the studies [33, 34] show that XAI methods often generate misleading feature importance. The observations [33] in different experiments show that unimportant features can be deemed to be relevant and important features to be irrelevant in SHAP. This is because XAI methods are highly dependent on data, which may produce spurious results [34]. Additionally, the explanations generated by XAI cannot be generalized. The proposed RAG-based explanations overcome this limitation of popular eXplainable AI (XAI) methods. The standard medical guidelines that clinical professionals endorse are provided to RAG, making the generated explanations more reliable than the results produced by XAI methods. Moreover, RAG-based explanations can be generalized for different datasets.

Most of the works, such as [4–7, 25], have used textual input to perform Q&A in clinical settings. However, these works have used limited information, leaving a vast amount of unutilized data. This is mainly due to the presence of information in different formats, such as images and tables, which is not suitable for LLMs. Moreover, the application of Q&A is more prominently displayed in literature rather than other downstream tasks regarding LLMs. Our work addresses these unexplored areas by leveraging multimodal data to perform a downstream task, namely, disease diagnosis with LLM. The research [28] is closest to our work, which is predicting diseases based on symptoms. The authors have used RAG to find similarities with medical literature, extract features, and perform disease classification with ML. The study in [28] uses only patient interaction data that could be incomplete and may generate erroneous results. Whereas, multimodal data (demographics, vitals, and radiological data) is being used in the proposed framework to provide comprehensive, relevant information for the LLM model to generate precise results.

Additionally, the existing LLM-based disease diagnosis studies such as [28–31] use ChatGPT, Gemini, and GPT-3.5, which have 175 B, 1.6 T, and 1.3 B parameters, respectively. They require a large amount of memory and computational power to store and perform any task. The storage and use of LLM-based medical applications in local settings, such as hospitals, is a challenging task because of high memory and computational requirements. Hence, this paper uses QLoRA in the framework as it is a suitable PEFT technique where memory requirement is a major challenge, as in the case of LLMs. Moreover, QLoRA performs better than its counter techniques, such as LoRA and adapters, because QLoRA has been developed upon them [35]. It has the advantages of LoRA and adapters, with quantization as an additional benefit.

The proposed work also generates explanations for diagnosis, which was not included in other comparative studies [28–31]. We have used RAG with LLM to generate explanations and reduce the hallucinations. We have opted for RAG instead of other DL techniques, such as contrastive learning, because DL-based techniques need a large medical explanation corpus for training of an explanation-generating LLM. However, to the extent of our knowledge, there is a lack of openly available datasets of annotated diagnosis explanations. Additionally,

the XLR-KGDD-generated explanations have been evaluated by a physician. Hence, XLR-KGDD is a novel multimodal disease diagnosis framework that generates explanations using state-of-the-art techniques, namely, KG, LLM, and RAG, to the best of our knowledge. The following section describes the XLR-KGDD framework.

3 XLR-KGDD: LLM and RAG-based disease diagnosis framework with explanations

This study introduces a framework for diagnosing diseases using LLM and RAG. It also includes the explanations for the diagnosis based on the standard medical guidelines. The framework consists of three main components: the Patient Attributes to KG (A2K) Mapper, the QLoRA-based Fine-Tuner (Q-FT), and the RAG-based Explanation Generator (REx), as illustrated in Fig. 1.

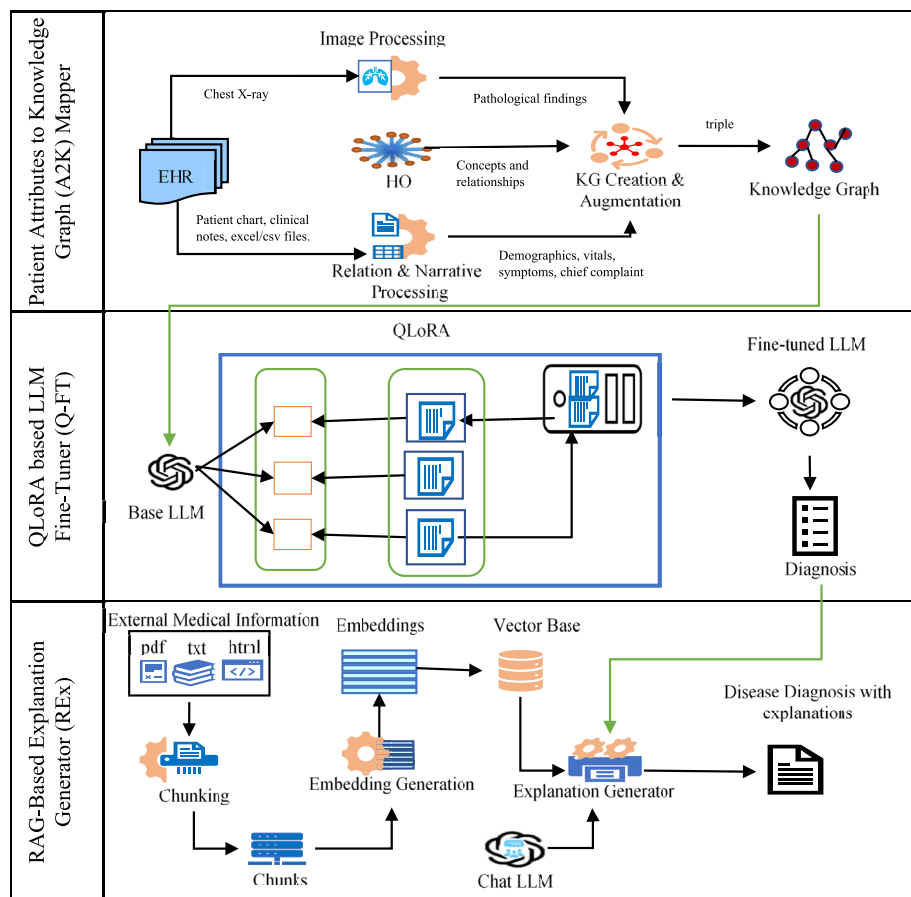


Fig. 1 Multimodal Disease Diagnosis Framework

Each component plays a significant role in the framework, starting from retrieving patient attributes stored in EHR to predicting disease diagnosis and producing explanations. The patient attributes are multimodal and stored in different formats such as radiological images are usually stored in '.dicom' format, clinical notes containing symptoms in '.txt', and demographics in relational tables ('.xlsx' or '.csv'). Therefore, the multimodal patient attributes are semantically mapped to a network, Patient KG, using the first component, A2K Mapper. It extracts relevant clinical attributes from EHR and maps them to a KG. The extraction process differs for each modality, and therefore, they are processed individually. To do so, A2K consists of two parallel processes: N2K mapper [36], which processes relational and textual data, and CheXzero [37], which processes radiological images.

Relation and Narrative Processing: The N2K mapper [36] is a Natural Language Processing (NLP)-based framework that maps patients' clinical notes to the Patient KG. The textual data in clinical narratives processed through chunking and Named Entity Recognition (NER) techniques. The important chunks are extracted from the narrative through a customized grammar proposed in [36]. NER has been applied to the extracted chunks and classified into symptoms, diseases, and pathological findings using standard biomedical ontologies. The chunks and their relevant classification are given as input to the module for the triple formation. Similarly, from the relational data such as demographics, the table attributes and table values are given as input to the *KG Creation and Augmentation* module.

Image Processing: The image processing block takes chest X-ray images as input and a trained DL, namely CheXzero [37]. The CheXzero is trained by a self-supervised contrastive learning technique to analyze chest X-ray images and identify the pathological findings present in images. It employs a vision transformer to extract features from the images and a text encoder to map these features to pathological findings. The CheXzero uses chest X-ray images and radiology reports for the training part and images for the prediction. It performs multilabel classification where the pathological findings are labels, and the model predicts the presence or absence of findings. The predictions are given to the *KG Creation and Augmentation* module, for triple formation.

KG Creation and Augmentation: KG is a set of triples. A triple is constituted of three parts: subject, predicate, and object. The subject and object are the entities, and the predicate is a relationship between the subject and object. The triple entities are obtained from the *Image processing* and *Relation and Narrative processing* modules. The predicate of a triple is extracted from the Healthcare Ontology (HO) [38], which stores the concepts (representing entities) and relationships between the concepts. These triples collectively construct a Patient KG. Further, Prompt engineering is applied to convert the Patient KG triples into a disease prompt (input for disease diagnosis), a suitable format for the LLM. The disease prompts serve as an input to the second component, Q-FT.

The Q-FT component fine-tunes the LLM for disease diagnosis by using the disease prompt as input. LLMs are trained on vast amounts of data and have billions of parameters. Hence, they need an extensive amount of memory with high computational capacity hardware, such as a Graphical Processing Unit (GPU) or TPU for loading and training. For instance, the LLaMA variant with 65 B parameters requires 780 GB of GPU Random Access Memory (RAM), which is equivalent to 16 A40 GPUs. Q-FT uses a PEFT technique, which offers several advantages in terms of memory and hardware requirements. Updating a large number of parameters, especially with a small dataset, easily leads to overfitting during the training or fine-tuning of a large model. On the other hand, PEFT updates only a small number of parameters while keeping most of the pre-trained weights fixed. This approach overcomes

the overfitting problem with small datasets while maintaining the high performance of LLMs. With the significant reduction in the number of updating parameters, the memory requirement also reduces substantially.

However, in a very large model setup, a small subset of parameters still constitutes a vast number. Hence, the proposed framework utilizes the Quantized Low-Rank Adaptation (QLoRA) [35], a PEFT technique, to fine-tune an LLM for disease prediction. QLoRA reduces the memory requirement by using quantization and converting the base LLM to a 4-bit precision model. QLoRA introduces 16-bit adaptors as additional parameters that are fine-tuned while keeping the rest of the pre-trained weights intact. QLoRA uses a 4-bit NormalFloat (NF4) datatype to minimize computation memory requirements. After loading the 4-bit precision LLM with the adaptors, the model is fine-tuned with disease prompts.

The LLMs are trained on a large dataset and store this information as weights. The dataset contains information from various sources, such as the internet and literature etc., covering different domains. However, LLM may produce inaccurate outputs in some instances due to hallucinations [13], making it unreliable to generate inferences for domain-specific queries such as medical diagnosis. The RAG-based Explanation Generator (REx) generates plausible explanations for the predicted diagnosis by leveraging LLM and external domain knowledge. In REx, we provide standard medical guidelines [39–42] as external domain knowledge to the LLM to overcome hallucinations and ensure the generation of relevant and sound inferences. REx accepts external domain knowledge in varied formats, such as Portable Document Format (PDF) or Text File Documents (TXT). These documents are split into chunks, and embeddings are generated for each chunk. These embeddings are stored in a vector database to eliminate redundancy in documents.

The explanation generator generates explanations for the diagnosis. It takes input from LLM and vector database. The input query embeddings are generated and compared with the vector database. The similarity between input query embeddings and vector database embeddings is computed to extract the ‘k’ most similar chunks of the external medical information. LLM uses extracted similar chunks to generate explanations for the given clinical decision. Since the LLMs are trained on a vast textual dataset, they generate reasoning for the diagnosis in natural language. The process of chunking, generating embeddings, and computing similarity used in this work is explained in the experimental study (Sect. 4). XLR-KGDD generates interpretable explanations based on standard medical guidelines and paves the way for reliable AI-generated clinical decisions. These explanations enhance the transparency and build trust between AI-based clinical decisions and physicians, which are usually the barriers to their acceptance.

The experimental setup for the proposed framework is described in the next section.

4 Experimental study

The Medical Information Mart for Intensive Care (MIMIC)-Eye [43] is a collection of various datasets such as MIMIC-CXR citation [44] (Chest X-ray images in.dicom format and semi-structured radiology reports in text), MIMIC-JPG [45] (Chest X-ray images in.jpg format and semi-structured radiology reports in text), MIMIC-IV [46] (attributes such as demographics, lab-events, ICD codes, diagnosis, symptom, chief complaint etc. in text and tabular format). It is the standard medical dataset usually used in multimodal studies [47, 48], which consists of various patient attributes in different formats, including images, radiology reports, symptoms, and demographics. Our system, based on the proposed framework XLR-KGDD, processes

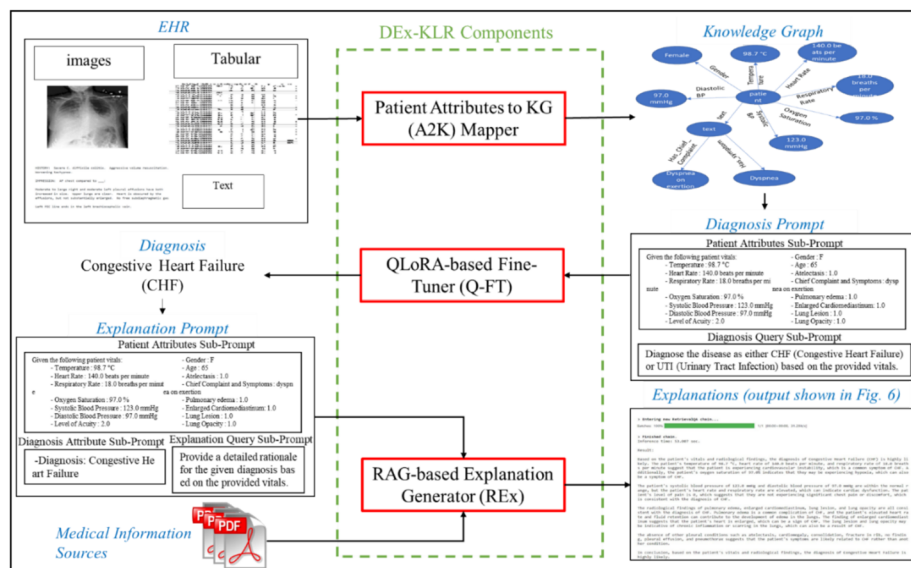


Fig. 2 The input and output of XLR-KGDD components

these multimodal patient attributes into a semantic network KG, using the first component, A2K Mapper. The input and output for each XLR-KGDD component during the experiment are illustrated in Fig. 2. The implementation of the proposed framework is carried out in Python language, and the Kaggle¹ platform is employed to obtain GPU support to fine-tune the LLM. Kaggle provides 16 GB GPU RAM and 29 GB CPU RAM for this purpose. A2K Mapper uses several Python libraries, including networkx, nltk, cv2, re, torch, transformers, peft, accelerate, bitsandbytes, trl and PyPDF. In A2K Mapper, the implementation of N2K Mapper and CheXzero is based on the papers [36, 37], respectively. N2K Mapper is used to map relational and textual data to KG, whereas CheXzero maps the medical images to KG.

The second component, Q-FT, fine-tunes the LLM model using the prompt (diagnosis prompt) generated from patient KG. Several prompts have been tried experimentally to select the most appropriate prompt. The openly available pre-trained LLMs are retrieved from the Hugging Face Portal² and Kaggle. Q-FT uses transformers, bitsandbytes and peft Python libraries to fine-tune three LLM variants, LLaMA-2, LLaMA-3, and Mistral, with QLoRA configuration which reduces the computing and memory requirements. The hyperparameters of QLoRA are tuned experimentally, and the following values are used to fine-tune the final models: the dimension of low-rank matrices is set to 16, the scaling factor for low-rank matrices is 8, and the dropout is 0.05.

The third component, REx, uses an explanation prompt, LLaMA-2 chat LLM, standard clinical guidelines including 2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure [39], ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure [40], the global cooperation barometer report [41] European Association of Urology Guidelines on Urological Infections [42] and various Python libraries, including langchain, chroma

¹ <https://www.kaggle.com/>.

² <https://www.huggingface.co/>.

DB, and transformers, to generate explanations. We have used the ‘RecursiveCharacterTextSplitter’ chunker to generate chunks from the standard clinical guidelines. It follows the policy of splitting and merging recursively while maintaining semantic integrity until every chunk is of the specified size. The embeddings for chunks are generated with a transformer-based model all-mpnet-base-v2.³ Each chunk is mapped to a 768-dimensional dense vector embedding. The similarity of the query prompt embedding vector and the embedded chunk vectors is calculated with the Maximal Marginal Relevance algorithm [49] and ranked according to their similarity. The most similar chunks, i.e., top-ranked chunks, are combined together by using chain_type (parameter of chroma DB vector database) with value ‘stuff’. The combined chunks are provided to the LLaMA-2 LLM along with the query prompt for explanation generation. The input and output for each XLR-KGDD component during the experiment are illustrated in Fig. 2.

This study considers two diseases, Congestive Heart failure (CHF) and Urinary Tract Infection (UTI), to test the feasibility of the proposed framework. However, it is a generic framework that can be extended further to diagnose any number of diseases. The performance of XLR-KGDD is evaluated using the following metrics: Accuracy, Precision, Recall, F1-score, Receiver Operating Characteristics (ROC) curve, and Precision-Recall (PR) curve [50, 51]. Random samples of the XLR-KGDD-based generated explanations have been taken and were evaluated by a physician. The obtained results are presented in the next section.

5 Results and discussion

This section discusses the results of various experiment settings used to evaluate the XLR-KGDD system. Three LLMs, namely, LLaMA-2, LLaMA-3, and Mistral, have been fine-tuned using the MIMIC-Eye dataset. The performance of each model is illustrated with two token lengths (128 and 512) paired with epochs (2 and 5). The obtained values of performance metrics, including accuracy, precision, recall, F1-score, AUC of ROC, and AUC of PR curves with LLaMA-2, LLaMA-3, and Mistral, are shown in this section. The comparison of the performances of fine-tuned LLMs helps to select the best model for disease predictions. Figures 3, 4 and 5 illustrate the ROC and PR curves obtained with the LLaMA-3, LLaMA-2, and Mistral models, respectively.

The AUC of ROC and PR curve shows an increasing trend in performance with an increase in token length and epochs. The AUC of ROC for CHF increased to 0.88 with a token length of 512 from 0.61 with a token length of 128 while keeping the same number of epochs, as shown in Fig. 3a, c. It indicates that increasing the token length improves the performance of LLMs as more information is being fed for training. On the other hand, with the same token length of 512, a comparatively moderate change is observed in the AUC of ROC for CHF from 0.82 to 0.88, as shown in Fig. 3c, e, showing that the model learning is complete, and it has been trained properly.

Figure 4 illustrates the results obtained with the LLaMA-2 model. In Fig. 4a–d, the same trend of increased AUC with longer token length has been observed. However, unlike LLaMA-3, there is no increment in the AUC of the ROC curve with increasing epochs in both cases of CHF and UTI, and there is even a decrease in the AUC of the PR curve for UTI, as shown in Fig. 4c–f. LLaMA-3 demonstrates stability in results with a continuous steady increment from 0.61 to 0.82 to 0.88 in the AUC of the ROC curve for CHF, as shown in Fig. 3a, c, e. On the other hand, LLaMA-2 shows a spike of 0.89 from 0.54 in the AUC

³ <https://huggingface.co/sentence-transformers/all-mpnet-base-v2>.

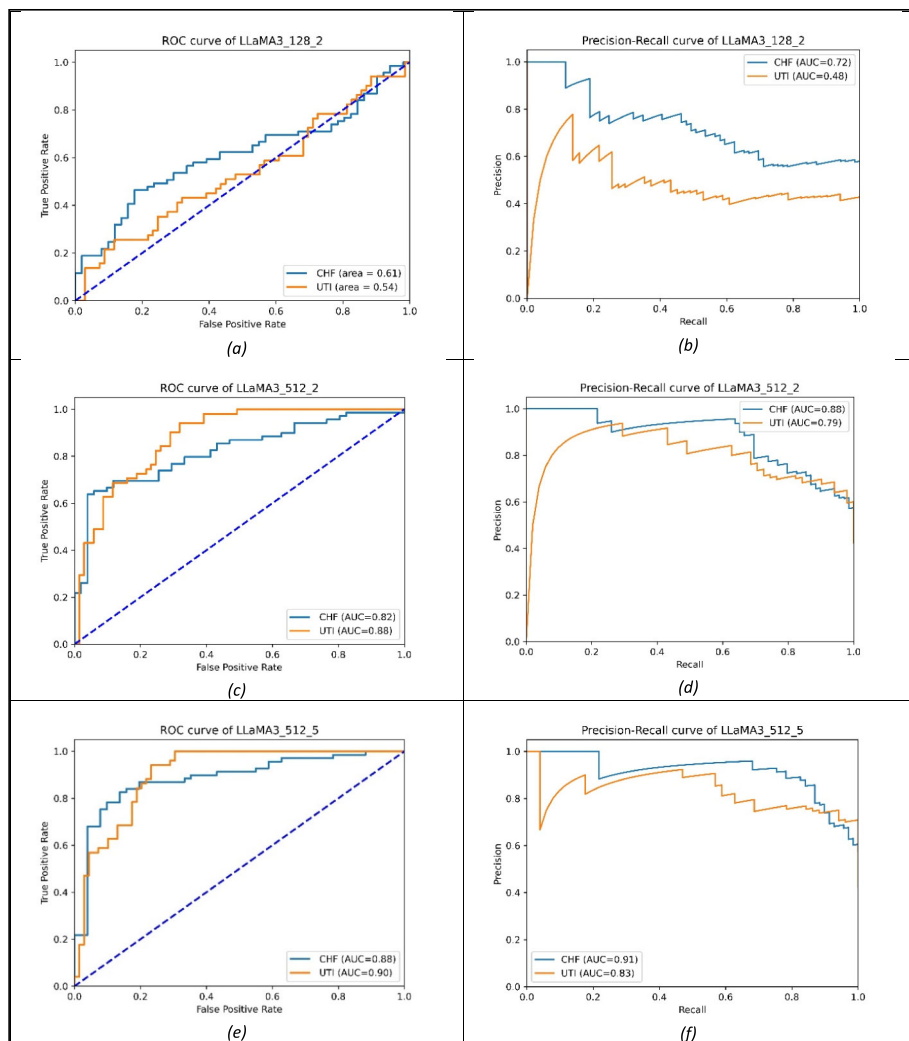


Fig. 3 ROC (a, c, e) and PR (b, d, f) curves with fine-tuned LLaMA-3 with 128, 512, 512 token length and 2, 2, 5 epochs, respectively

of the ROC curve and a spike of 0.91 from 0.60 in the AUC of the PR curve for CHF, as illustrated in Fig. 4a, c, e.

Figure 5 shows the result obtained with Mistral, which shows a peculiar behavior in the case of CHF. As the number of epochs increases, the AUC of ROC and PR curves decreases from 0.89 and 0.93 to 0.88 and 0.87, respectively, as shown in Fig. 5c, e. A similar trend is observed with LLaMA-2 for UTI. However, unlike LLaMA-2, Mistral depicts an increasing trend in AUC of ROC and PR curves for UTI from 0.85 and 0.75 to 0.87 and 0.80, respectively, for the same token length and epochs, as presented in Fig. 4d, f. Table 1 shows the values obtained by the LLMs in various performance measures, including Accuracy, Precision, Recall, F1-score, Area under the ROC curve (AUROC), and Area under the PR curve (AUPRC). LLaMA-3

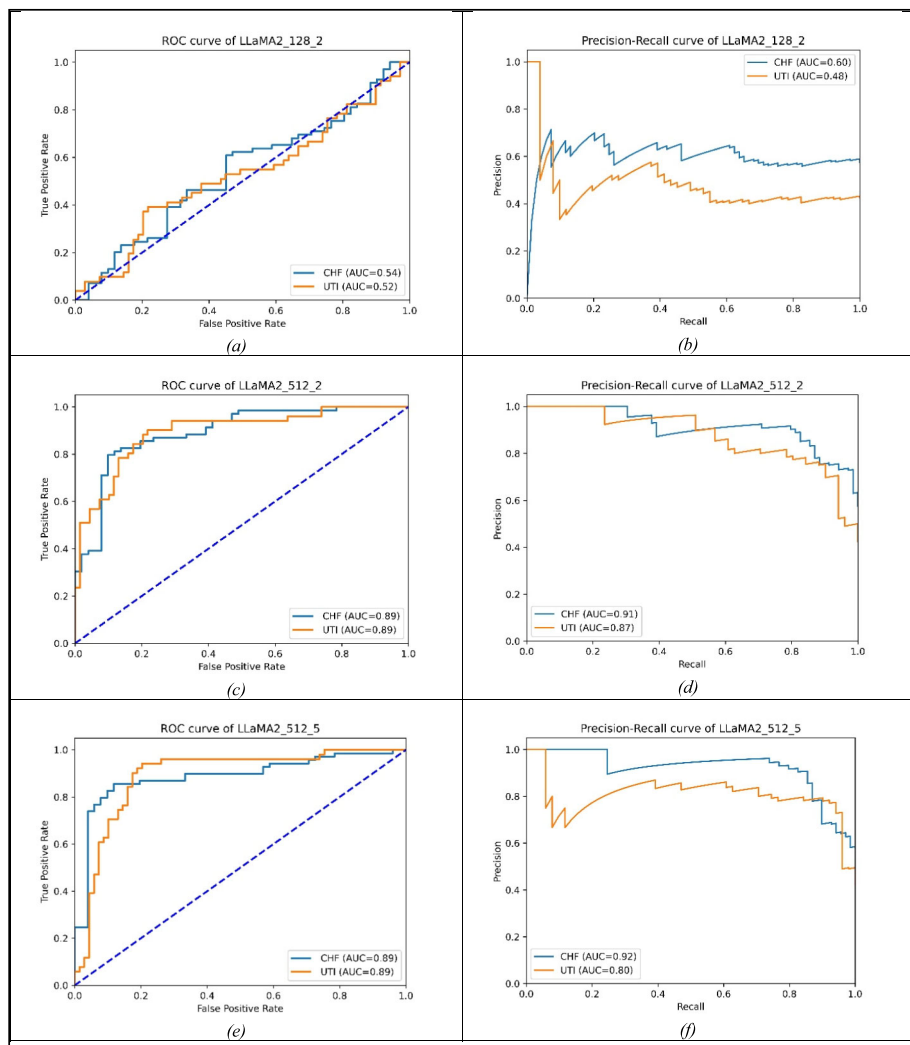


Fig. 4 ROC (a, c, e) and PR (b, d, f) curves with fine-tuned LLaMA-2 with 128, 512, 512 token length and 2, 2, 5 epochs, respectively

gives better performance in the AUC of the ROC curve and the AUC of the PR curve for UTI. It shows comparable performance with LLaMa-2 for CHF (as visible in the AUC of ROC and the AUC of the PR curve of CHF). However, LLaMA-3 shows slightly lower values when measuring the performance in accuracy, precision, recall, and F1-score. This could be because of the imbalanced data where the CHF class has more samples than UTI. The ROC curve and PR curve are more robust performance metrics as they utilize different thresholds to evaluate the AUC values and, hence, are able to handle imbalances in the dataset efficiently [52, 53]. On the other hand, accuracy, precision, recall, and F1-score use a fixed threshold for the confusion matrix, which makes them relatively more rigid metrics than robust metrics like ROC and PR curves.

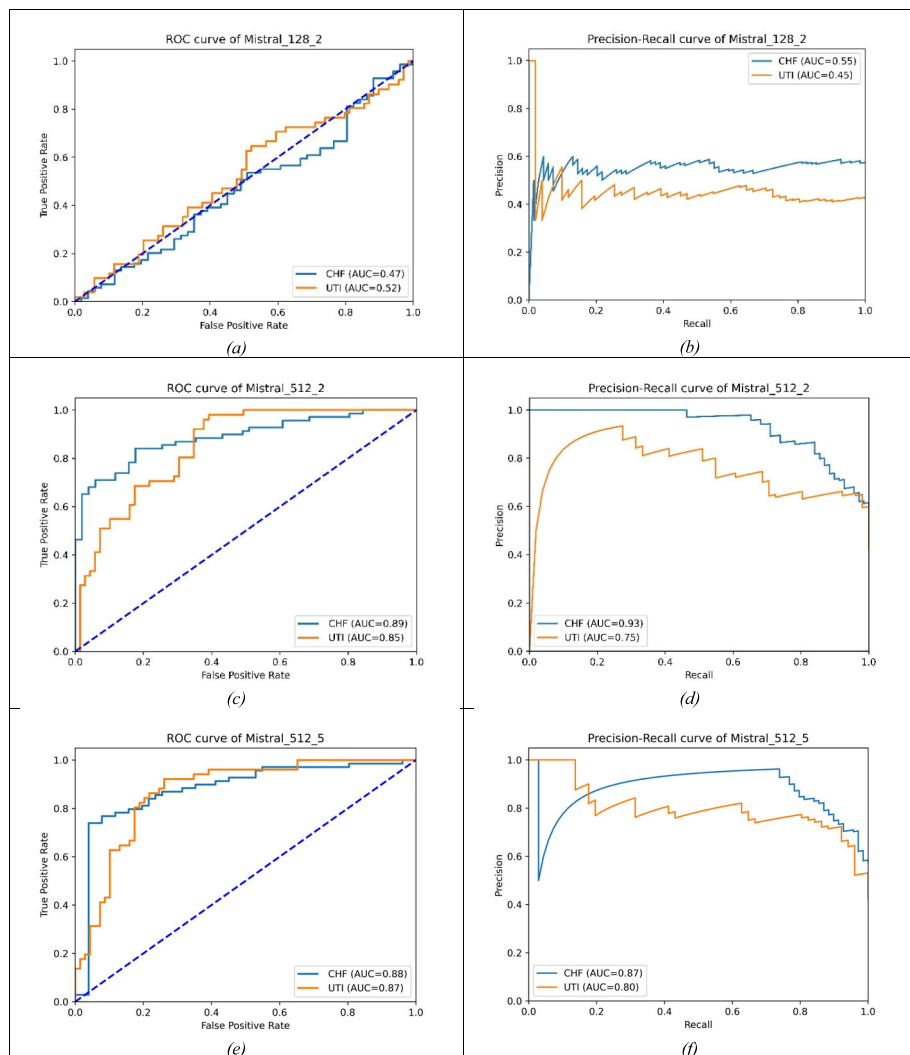


Fig. 5 ROC (a, c, e) and PR (b, d, f) curves with fine-tuned Mistral with 128, 512, 512 token Length and 2, 2, 5 epochs, respectively

In this study, QLoRA-based quantized LLMs are used, which reduces the precision from 32-bit to 4-bit. It reduces memory requirements but, at the same time, impacts the performance of LLMs. Hence, the proposed framework can demonstrate even better performance by providing more resources in terms of memory and GPU.

Results of RAG-based Explanation Generator (REx) Fig. 6 illustrates the explanations generated by the LLaMA-2 chat model with multimodal patient attributes as input. The explanations also indicate that the diagnosis from the XLR-KGDD is the most likely outcome, even when decisions are made based on standard clinical guidelines. The explanations for the clinical decision are described in terms of the given patient attributes and clinical knowledge.

Table 1 Performance Measures obtained with fine-tuned LLaMA-3, LLaMA-2 and Mistral

Models	Token Length	Epoch	Accuracy	Precision	Recall	F1-Score	CHF		UTI	
							AUROC	AUPRC	AUROC	AUPRC
LLaMA-3	128	2	0.59	0.58	0.59	0.58	0.61	0.72	0.54	0.48
	512	2	0.77	0.79	0.77	0.77	0.82	0.88	0.88	0.79
	512	5	0.83	0.84	0.83	0.83	0.88	0.91	0.9	0.83
LLaMA-2	128	2	0.52	0.53	0.52	0.52	0.54	0.6	0.52	0.48
	512	2	0.82	0.84	0.82	0.83	0.89	0.91	0.89	0.87
	512	5	0.87	0.88	0.87	0.87	0.89	0.92	0.89	0.8
Mistral	128	2	0.5	0.53	0.5	0.5	0.47	0.55	0.52	0.45
	512	2	0.78	0.79	0.78	0.78	0.89	0.93	0.85	0.75
	512	5	0.81	0.81	0.81	0.81	0.88	0.87	0.87	0.8

Bold values indicate the best results obtained in the experiment

```

> Entering new RetrievalQA chain...
Batches: 100%  1/1 [00:00<00:00, 31.29it/s]

> Finished chain.
Inference time: 53.007 sec.

Result:

Based on the patient's vitals and radiological findings, the diagnosis of Congestive Heart Failure (CHF) is highly likely. The patient's temperature of 98.7 °C, heart rate of 140.0 beats per minute, and respiratory rate of 18.0 breaths per minute suggest that the patient is experiencing cardiovascular instability, which is a common symptom of CHF. Additionally, the patient's oxygen saturation of 97.0% indicates that they may be experiencing hypoxia, which can also be a symptom of CHF.

The patient's systolic blood pressure of 123.0 mmHg and diastolic blood pressure of 97.0 mmHg are within the normal range, but the patient's heart rate and respiratory rate are elevated, which can indicate cardiac dysfunction. The patient's level of pain is 0, which suggests that they are not experiencing significant chest pain or discomfort, which is consistent with the diagnosis of CHF.

The radiological findings of pulmonary edema, enlarged cardiomeastinum, lung lesion, and lung opacity are all consistent with the diagnosis of CHF. Pulmonary edema is a common complication of CHF, and the patient's elevated heart rate and fluid retention can contribute to the development of edema in the lungs. The finding of enlarged cardiomeastinum suggests that the patient's heart is enlarged, which can be a sign of CHF. The lung lesion and lung opacity may be indicative of chronic inflammation or scarring in the lungs, which can also be a result of CHF.

The absence of other pleural conditions such as atelectasis, cardiomegaly, consolidation, fracture in rib, no finding, pleural effusion, and pneumothorax suggests that the patient's symptoms are likely related to CHF rather than another condition.

In conclusion, based on the patient's vitals and radiological findings, the diagnosis of Congestive Heart Failure is highly likely.

```

Fig. 6 Explanation Generated by REX with LLaMA2 model

For example, the explanations suggest that the patient's heart rate (140 beats per minute) and respiratory rate (18 breaths per minute) are elevated, which is a sign of CHF. According to clinical guidelines, the normal range of heart rate for a healthy human is between 60 to 100 beats per minute, and the normal range of respiratory rate is between 12 to 16 breaths per minute.⁴ Both of the patient's vital signs are on the higher side than normal, which is consistent with the REX generator.

The radiological findings also play an important role in clinical decisions, as shown in the explanations generated by the REX. Pulmonary edema and an enlarged cardio-mediastinum are possible indicators of CHF. Their significance and physiological impact are explained in simple terms which are understandable to humans. In the final statement, the REX-generated explanations support the diagnosis with evidence from clinical guidelines. The explanations have been evaluated by a clinician and are found to be relevant.

Discussion Although we did not come across completely similar studies in the literature in terms of multimodality and LLM, we have compared our work with other LLM-based disease diagnosis studies. Table 2 shows the comparison of existing literature on disease diagnosis using LLM and the proposed XLR-KGDD.

In [31], the Gemini model has a very high precision value but a very low recall value. This suggests that the Gemini model is possibly overfitting for positive instances and performing well in finding positive cases. However, it struggles to identify true positives and does not generalize well to the dataset. On the other hand, the proposed model does not appear to be overfitting, as its recall and precision values are closer together. Moreover, none of the studies provide explanations for the decision, whereas XLR-KGDD also gives clinicians interpretable explanations in addition to disease diagnosis.

⁴ <https://www.hopkinsmedicine.org/>.

Table 2 Comparison with existing LLM-based disease diagnosis studies

Research Works	Accuracy	Precision	F1-score	Recall
Health-LLM [28]	0.83	×	0.76	×
Gemini Model [31]	×	0.97	0.80	0.69
GPT-3.5 [31]	×	0.90	0.86	0.84
XLR-KGDD (proposed work)	0.87	0.88	0.87	0.87

The interpretation of the results by addressing the challenges in the disease diagnosis is summarized below:

- *Processing of multimodal data:* The clinical data is inherently multimodal as it contains varied types of information, including demographics, symptoms, radiology images (such as X-rays), radiology reports, and vitals readings. Each type of information plays a significant role in disease diagnosis. The framework uses KG to semantically integrate the varied types of clinical data into a hierarchical structure with the help of Healthcare Ontology (as illustrated in Fig. 1 of Sect. 3). It enriches the hierarchical structure with semantic information, such as (symptoms and lab-events) in the form of relationship labels. Moreover, KG is scalable and able to incorporate other modalities in the future without disrupting the existing information.
- *High memory computation:* LLMs have a high number of parameters and, therefore, require a lot of computation power and time to train or fine-tune the model (as mentioned in Sects. 2 and 3). In the medical domain, frequent fine-tuning with new data is necessary, but the high computation cost and time make it infeasible to do frequent fine-tuning for the system. Hence, the framework used PEFT and QLoRA to quantize the model before fine-tuning, which significantly reduced the training time and computation cost. For instance, the LLaMA 2 model has 7B parameters that are stored in 14 GB of memory, whereas the quantized model takes only one-fourth of the original size.
- *Hallucination:* The LLMs can be trained to generate medical explanations for diagnosis, but it requires a large number of instances (explanations) validated by doctors. However, the absence of such data makes it hard to provide medically relevant explanations for diagnosis as LLMs are highly susceptible to hallucinations (as described in Sect. 3) in complex and knowledge-intensive problems (such as explanation generation) with a low amount of training data. Hence, the framework uses the RAG technique, which provides the medical guidelines (validated by doctors worldwide) as additional context to the LLMs. It reduces the hallucinations and generates relevant responses as explanations.
- *Human interpretable explanations:* In the medical domain, the adaptation of AI-based applications, especially in disease diagnosis, depends on the trust of physicians and AI methods. However, the black-box nature of AI methods is a hindrance to the establishment of that trust. XAI methods tried to overcome this difficulty by introducing explanations. Still, they also require a domain expert to interpret these explanations as they are present in graphical format rather than the natural language understandable by physicians (as mentioned in Sect. 2). We have addressed this challenge by generating physicians' understandable and relatable explanations in natural language. Moreover, the generated explanations are more relatable to physicians as standard medical guidelines have been provided as additional context to LLM.

Limitations: The XLR-KGDD framework addresses many problems, such as disease diagnosis with explanation, multimodal clinical data, and hallucination in explanations. However, there are a few limitations, which are given below. Firstly, we have used quantization (QLoRA) to reduce the memory requirement, which may affect the performance of the trained model to some extent. Secondly, due to the imbalanced dataset, some biases might be introduced during the training phase. Finally, it was observed that a few statements from explanations have discrepancies, such as inconsistencies in the range of vitals and vagueness in the sentences in some places. For example, “normal temperature” was written as “elevated”.

6 Conclusion and future work

Large Language Models (LLMs) such as ChatGPT have recently gained public attention due to their capability to generate responses like humans. Existing studies have primarily focused on using LLMs for question-answering or information-extraction tasks. However, their potential in different domain applications with multimodal data, such as disease diagnosis, is a relatively unexplored research area. Our paper has addressed these research gaps by proposing an LLM and RAG-based explainable disease diagnosis framework, XLR-KGDD, with multimodal patient clinical information. The framework consists of three components: the Patient Attributes to Knowledge Graph (A2K) Mapper, the QLoRA-based Fine-Tuner (Q-FT), and the RAG-based Explanation Generator (REx). The A2K Mapper organized the multimodal patient clinical information into a semantic network, Knowledge Graph (KG), which semantically connects each patient detail. The Q-FT component fine-tuned the base LLMs for disease diagnosis using a Parameter Efficient Fine-tuning (PEFT) technique, namely QLoRA. It substantially reduced the memory and high computational requirements during fine-tuning. The REx component was finally employed to generate explanations for the diagnosis using the Retrieval Augmented Generation (RAG) technique. REx used the standard medical guidelines as an external knowledge source to overcome hallucination problems in LLMs. XLR-KGDD has been tested on a multimodal Electronic Health Record dataset, MIMIC-Eye. Three popular and Open-source LLMs, LLaMA-2, LLaMA-3, and Mistral, were fine-tuned for disease diagnosis. The results illustrated that with the same configuration of token length of 512 and 5 epochs, LLaMA-3 gave more robust results than its peers in the AUC of the ROC curve and AUC of the PR curve, with values of 0.88 and 0.91 in diagnosing Congestive Heart Failure and 0.90 and 0.83 in diagnosing Urinary Tract Infection, respectively.

In the future, we aim to enhance our framework's performance by applying data imbalance mitigation techniques and investigating the impact of various quantization techniques on diagnosis accuracy. We also plan to address and resolve the limitations in explanation (vagueness and inconsistency) by introducing more appropriate and inclusive guidelines to LLM and training the LLM on a large medical corpus to generate medically precise explanations.

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Author contributions S.D. has contributed in conceptualization, data curation, system development, result analysis, and writing—original draft. A.T. and P.B. has contributed in conceptualization, supervision, result analysis, and writing—review & editing.

Data availability The data that support the findings of this study are available from the paper titled “MIMIC-Eye: Integrating MIMIC Datasets with REFLACX and Eye Gaze for Multimodal Deep Learning Applications” (<https://doi.org/10.13026/pc72-as03>).

Declarations

Conflict of interest The authors declare no competing interests.

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